

P8109 Spring 2026 Homework #2

Due on February 26 at 11:59pm

1. Let $X_1, \dots, X_n$ be i.i.d. $\text{Poisson}(\lambda)$ random variables. Find the MLE of λ .
2. Consider a sequence of i.i.d. random variables from a $N(\mu, \sigma^2)$ distribution. Let

$$m = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

- (a) Given that $\sigma^2 = \mu$, show that the MLE of μ is

$$\hat{\mu} = \frac{-1 + \sqrt{1 + 4m}}{2}.$$

- (b) Given that $\sigma = \mu$, show that the MLE of μ is

$$\hat{\mu} = \frac{-\bar{X} + \sqrt{\bar{X}^2 + 4m}}{2}.$$

3. Let $X_1, \dots, X_n$ be a random sample from a $N(\mu, 1)$ distribution.

- (a) Show that

$$T = \frac{1}{n} \sum_{i=1}^n X_i^2 - 1$$

is an unbiased estimator of μ^2 .

- (b) Explain why $2T$ is not an appropriate estimator of $2\mu^2$.
 - (c) Give a simple proof that T is a biased estimator of μ .
4. (a) Define what it means for T_n to be a consistent estimator of a parameter θ .
(b) Given T_n is unbiased for θ , use Chebyshev's inequality to show that T_n is consistent for θ if

$$\text{Var}(T_n) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

5. Let $X_1, \dots, X_n$ be i.i.d. $N(\theta, 1)$ random variables.

- (a) Obtain the CRLB for any unbiased estimator of θ .
- (b) By writing the score function in suitable form, identify the function $\tau(\theta)$ for which an MVBUE estimator exists. Write down the latter estimator.
- (c) Write down the UMVUE for θ .
- (d) Obtain the CRLB for any unbiased estimator of 2θ .

- (e) By writing the score function in suitable form, identify the MVBU estimator of 2θ if it exists.
6. Let $X_1, \dots, X_n$ be i.i.d. $\text{Poisson}(\lambda)$ random variables. We wish to estimate $p = P(X_j = 0)$.
- (a) Find an expression for p in terms of λ .
- (b) Find the CRLB in terms of p for any unbiased estimator of p .
- (c) Let

$$Y_j = \begin{cases} 1, & \text{if } X_j = 0, \\ 0, & \text{otherwise.} \end{cases}$$

- Find EY_j and use this result to write down an unbiased estimator T of p .
- (d) Find $\text{Var}(T)$ (in terms of p).
- (e) By graphing suitable functions or otherwise, determine if T is the MVBU estimator.